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A Quantum-Driven Trust-Aware Optimization Framework for Secure and Energy-Efficient Healthcare IoT Networks

Abstract- The explosive growth of IoT devices in the healthcare system increases the pressure over the issues of data security, trust management and energy utilization. Typical cryptographic and clustering approaches fall short as they are unable to effectively tackle dynamic threats like DDoS attacks and node-level compromises. In this study, a QIoT framework is proposed for the improved security, trust and energy-saving healthcare data communication in WSNs. The resultant model Fuses Quantum Key Distribution (QKD), Trust aware Mountain Gazelle Optimization (TMGO) and dynamic trust based on residual energy for intelligent CH selection in WSN. The system further includes a quantum-inspired anomalous detection function acting to detect and mitigate DDoS attacks in real-time, thereby enabling secure uninterrupted communication of data throughout the network. Simulations conducted in MATLAB for WSN with 100-nodes show the better performance of our proposed (QIoT-TMGO) as compared to benchmark approaches: LEACHTM, DGTSSA and eeTMFOGA. The model increases network stability by 24%, raises the average trust up to 32% and maintains over 95% data delivery rate under 10% malicious nodes. It also significantly improves energy-efficiency metrics including FND, HND and LND, which indicates that it is robust and suitable for secure and sustainable healthcare systems.

Keywords- Quantum IoT, Wireless Sensor Networks, Healthcare Data Security, Quantum Key Distribution, Trust-aware Clustering, Meta-Heuristic Optimization, DDoS Mitigation, Energy Efficiency.

I Introduction

The incorporation of Internet of Things technologies within healthcare infrastructure has provided aid in real-time patient monitoring, decision-making data in treatment, and the management of chronic conditions. The remote sensing devices, wearable monitor & ambient diagnostic platforms are the integrated wireless ecosystem that monitors and transmits health-related data necessary for accurate prediction decisions about patients. However, this network expansion poses some challenges, especially in data privacy issues, node reliability, and energy efficiency [1],[2],[3]. Nevertheless, existing frameworks based on either traditional cryptographic models or lightweight routing protocols are more or less ill-suited to be prepared against adversarial behaviors or coordinated attacks [4],[5]. Furthermore, healthcare IoT deployments usually (most if not all of the time) concern highly resource-constrained environments where conventional security overheads cannot be supported [6], [7]. Dimensioning cluster mechanisms disregard node trustworthiness [8], [9], leading to windows of opportunity for inside compromises, and reactive security systems are too late for life-critical medical applications [10],[11]. In situations such as remote care or emergency services, trust-based communication and secure data aggregation are indispensable. However, many existing models only study the latter type of behaviors in which seemingly cooperative nodes can disrupt CH elections and change data flows [12],[13].

Although a variety of protocols serve for energy-aware routing or secure key exchange alone, few can combine these issues in one responsive architecture [14], [15], [16]. In healthcare scenarios, we believe that applications are prone to highly spread (e.g., virus)-distributed denial-of-service (DDoS) attacks which can be generated from infected but apparently operational nodes, making reactive anomaly detection ineffective [17], [18]. Trust metrics employed in the decision making of routing rely on fixed thresholds or lagged behaviour analysis which all makes it incapable of finding

coordinated or adaptive threats [19]. Without DQKA, the real-time QRSAsU8 security mechanism of healthcare networks is lacking and subject to spoofing/replay attacks as in particular those taking place at CH level with data fusion [20], [21]. The Quantum Key Distribution (QKD) is based on the no-cloning theorem and quantum entanglement, which can be secure against a quantum-capable adversary who can do nothing for it [14], such that we know how to prove [21]. On the other hand integrating QKD in WSN was not attempted much because of its problem overhead, scalability and integration complexity [22], [23]. The QKD is combined with a trust-aware clustering mechanism, managed by an on-demand meta-heuristic optimization technique that enables making such decision adaptive and responsive to balance between trust decaying r and the remaining energy at node n [24], [25] thus ensuring secure routing activities in real-time while under adversarial environments, without trading off the life of the system or reducing its scalability.

It is obvious that this potential can only be realized if combined with a comprehensive architecture in which secure KE, adaptive trust modeling, and energy-efficient CH selection are effectively integrated. Unlike typical static security policies that assume nodes are stable over time, frequent reevaluation of trust is needed in healthcare environments because usage patterns change over time; the quantity of health data changes; and the extent of anomalies at node level increases. In reality, the employed attack is to exploit remaining energy metrics by adversaries and attempt to fulfill CH that can be eliminated in case a recent mechanism (behavioral trust analytics) is incorporated into the optimization model [3], [9], [13]. In this paper, we introduce the Trust-aware Mountain Gazelle Optimization (TMGO) algorithm and apply it for multi-objectives CH selection using trust scores obtained from direct and indirect monitoring results along with energy levels as well as QKD handshake success rates. To enrich the DF-CUDA system, we introduce a new quantum-inspired anomaly detection mechanism that focuses on describing DDoS signatures when the quantum key lifetime is updated and when a message travels indefinitely before it is received. In tandem, this will allow the network to natively preempt rogue nodes, reroute sensitive data, and save energy. The impact of end-to-end analysis results: Our proposed QIoT system provides greater network security, trusted estimation and data communication efficiency in hostile environments, which is verified by the experiment conducted by us in healthcare examples [7], [15], [17], [21]. The paper makes three main contributions:

- Towards a hybrid QIoT architecture by using Quantum Key Distribution and trust aware cluster head selection for enhanced data confidentiality, integrity, and energy efficiency in health care sensor networks.
- The Quantum -inspired Anomaly Based DDoS detection mechanism that can be used for a real-time proactive detection and mitigation of the DDoS attack and is specifically designed to support the resource constrained medical devices.
- Development of dynamic and secure cluster formation algorithm with trust metrics based on multi-parametric inputs such as energy levels, etc., termed as Trust-aware Mountain Gazelle Optimization (TMGO).
- Extensive simulation and MATLAB-based evaluation demonstrate that the proposed scheme significantly outperforms LEACH-TM, DGTSSA, and eeTMFOGA on trust accuracy, delivery reliability and energy consumption metrics.

The remainder of the paper is structured as follows. The related works section reviews prior efforts in trust-based clustering and secure healthcare IoT. The methodology outlines the proposed QIoT-TMGO framework and its core components. Experimental results and comparisons are presented, followed by the conclusion and future directions.

II Related Works

For secure transmission of biomedical images, Kaur et al. (2021) developed a compressive sensing-based hybrid encryption method with the use of a five-dimensional hyper-chaotic map (FDHC). A θ -NSGA-III algorithm was introduced with an optimized chaotic system to form dynamic keys, resulting in a sensitivity and encryption speed improvement for biomedical IoMT systems [1]. The tuning mechanism of this approach supported image-level security that is crucial for resource-limited healthcare settings. Saba et al. tried for lower communication costs. Balancing energy load among biosensor nodes is an important consideration for SEF-IoMT (2020). Their design includes network-aware optimization and lightweight encryption to protect patient data from being accessed without permission [2]. In simulated tests, the model magnified improvements in throughput and drop/latency. In this context, Mondal et al. proposed a solution to cater to the energy constraints in biosensor transmissions. It showed a secure health monitoring with an encrypted CBC-based encryption using lightweight facilities. Their work improved network lifetime by improving lossless transmission and data integrity [3]. The solution provided real-time security by using the protocol with less communication overhead.

Highlighting the vulnerability of wearable medical devices, Jan et al. For secure pairing and authentication, LightIoT (2021) is a lightweight communication scheme. It created energy-efficient, secure sessions with wearables and remote servers for defense against adversarial threats [4]. The three-phase handshake allowed for both data privacy and computational feasibility. Sharma et al., focusing on energy conservation, In the year 2019, Ref proposed a model for psychological and neurological disorder monitoring by using IoT in the healthcare system. Their approach was focused on designing specialist clinical hardware for minimum power and, like us, targeted specific clinical applications [5]. This then set the foundation for context-aware IoT deployment in mental health. To better the CH selection, Kapoor et al. Tong et al., in 2023, proposed the SS-ROW algorithm, which integrated squirrel/sparrow swarm intelligence and ciphertext-policy encryption. Therefore, both trust and inter-cluster distance were optimized with energy to secure routing and keep the sensors always alive in the remote patient monitoring system [6]. Their combination of hybrid encryption made it compelling enough to keep the security tight and data confidentiality bold.

In the meantime, Balakrishnan et al.(2024) suggested human-to-patient transmission of MDV in a highly homologous host environment. They proposed an EMCQLR cluster with elliptic curve cryptography into a secure healthcare IoT framework. After the device data routing was optimized using reinforcement learning routing [7], double hash biometrics were used to authenticate and filter signals, securing data passing from node to node. This had the effect of protecting health systems against real-time privacy-preserving analytics. Chakraborty, Maji, et al. employed fog computing in their work to provide contextual modality recognition using wearable sensors. A secure data aggregation approach has been proposed by (2024) in the form of FC-SEEDA, where IoT nodes are relieved from transmission. Decentralized architecture made it more scalable, and built-in cryptographic mechanisms ensured security during the edge-to-cloud transitions [8]. This approach addressed both energy efficiency and low-latency data flows.

Evolution in cloud-integrated healthcare has been the focus of study by Ahamed et al.(2023) presented EE-HDMM, which is a resource allocation system that makes extensive use of wearable sensors and provides cloud analytics. The model also resulted in minimized communication delays combined with energy efficiency as a result of optimal sensing cycles [9]. It had good sensitivity and precision accompanied by high performance in terms of evaluation metrics, making it ideal for critical monitoring. Singh et al. (2022) proposed OptiGeA, a routing protocol for IoMT that selected CHs

using node heterogeneity and energy proximity. A very recent study has shown that this strategy of sinks, in the form of mobile units paired with tailored utility functions, eliminated transmission bottlenecks and prolonged operational life under disease surveillance [10]. Their system remained robust under the epidemic big data monitoring scenario. To improve the device identification, Kumar et al. (2024) proposed derivatives based on Physical Unclonable Functions (PUF), using a blockchain-based approach and an introduced Proof-of-Energy consensus. It was targeted at tamper-resistant identity management combined with energy savings in decentralized healthcare networks [11]. The scalability of the former is in line with IoMD deployments for Healthcare 4.0. A remarkable scaling protocol that correctly accounts for synchronization when appropriate effects are turned off. Mahendran et al. gave a solution for energy bottlenecks in IoT sensors. To minimize this idle time, we optimized the sensor activity cycles through the CDTs scheduling protocol in (2021). The error and delay were also minimized by bit interleaving with ECC encryption for medical data transmissions [12]. This species had great grounds to be well adapted for the real-time monitoring of physiological activities. It helps in solving inflexible encryption routines, addressing Khashan et al. Introduced FlexCrypt, a lightweight solution, selecting the encryption parameters dynamically based on the real-time energy status of nodes (2021); it had in turn added a mobility-assisted clustering algorithm and a key management subpart to provide secured and configurable communication [13].

To circumvent traditional roadblocks in the diagnosis, Aggarwal et al. (2023) predicted future quantum computing where high-fidelity granular precision was modeled using the Qubit-level operation, probabilistic paring, and biomolecular simulation. The model decreased time to diagnosis and improved patient-level as well as deep level-diagnosis [14]. Dynamo-net, Verma and Jha presented an energy-aware routing mechanism by combining mutation based multilayer perceptron and BSP-GSA (binary sequential paradigm grey wolf optimization) for CH optimization to get the best path in dynamic networks. They embedded the user-level self-signature and protocol-level elliptic curve encryption of communication security in their scheme for secure transmissions [15]. Gupta et al., Overcoming the design challenge in densely deployed IoT networks (2023), an EER-RL reinforcement learning-route LPWAN specific to healthcare application is used. Theirs is more scalable and routing efficient in adjusting the dynamic variations of node density and energy parameters [16]. To reduce costs of transmission and ensure privacy of medical history, Ganesh Prabhu et al. (2024) suggested a function-preserving encoding and established IoMT-based secure power-efficient framework for the latter to facilitate safe patient checking. In body area networks [17], it reduces the power consumptions and protects data sent to untrusted or malicious base stations. The confidentiality of files has been enhanced.

Table 1: Comparative View of Existing Security and Optimization Strategies in Healthcare IoT

Contributor	Quantum	Trust	Energy	Detection	Strategy	Limitations Identified
Kaur et al. [1]	✗	✓	✓	✗	Trust-based clustering	No integration of quantum security
Saba et al. [2]	✗	✗	✓	✓	AI-enabled energy routing	Lacks quantum and trust metrics synergy
Mondal et al. [3]	✗	✓	✓	✗	Secure transmission	No real-time anomaly detection

Verma & Jha [15]	✗	✓	✓	✓	MUMLP, IECC	Limited scalability in dense networks
Aggarwal et al. [14]	✓	✗	✗	✗	Precision-Granular QC	Only theoretical, not IoT-integrated
Gupta et al. [16]	✗	✗	✓	✗	RL-based routing	Security overlooked
Ganesh Prabhu et al. [17]	✗	✓	✓	✓	Secure IoMT framework	No quantum encryption or trust optimization
Proposed QIoT-TMGO	✓	✓	✓	✓	QKD + TMGO + Anomaly Detection	Comprehensive multi-layer optimization

Table 1 presents a comparative analysis of key existing frameworks in healthcare IoT, focusing on quantum integration, trust-awareness, energy efficiency, and detection capabilities. It highlights the limitations of prior works in addressing all these aspects simultaneously. The proposed QIoT-TMGO framework distinctly outperforms by offering a holistic, quantum-secured, and trust-optimized solution.

2.1 Research Gap

Despite the developments in secure IoT-empowered healthcare platforms, today's models still do not inherently accommodate the dynamic adaptability to resource constraints at runtime encryption. Existing trust-aware clustering methods usually do not consider energy consumption as well as uncertainty of behaviors in the harsh environment. There are few practical wireless sensor network architectures that include quantum level security features such as QKD. Most of the energy efficient routing approaches do not consider malicious node detection at time of data aggregation. The machine learning-based and reinforcement methods enhance routing, but not much resilient to DDoS-like traffic pattern. Most existing solutions also consider security, trust, and energy management separately. However, an involved quantum-secure trust-based optimization framework is still significantly unexplored for healthcare IoT networks.

III Proposed Work

The proposed work introduces a quantum trust system along with an energy-efficient secure architecture for healthcare devices in IoT. Integral to this architecture is the incorporation of Quantum Key Distribution (QKD), whereby secure cryptographic keys are generated using quantum mechanics, making them invulnerable against eavesdropping or key compromise even against a quantum computer. This paper introduces a Trust-aware Mountain Gazelle Optimization (TMGO) algorithm to find the optimal cluster head that balances between energy management and trust at the same time in light of residual energy, communication reliability, and past behavior. In contrast to a static clustering, this mechanism is able to adjust according to network dynamics and malicious behaviors. Inspired by existing initiatives that have previously prioritized privacy-preserving lightweight designs in medical IoT [33], the framework improves data routing by focusing on nodes that have high trust scores while at the same time depleting less energy. The clustering phase also reduces communication overhead, leading to a longer sustainable IoMT routing lifetime [26],[27]. Additionally, a lightweight anomaly

detection layer inspects traffic for Distributed Denial of Service (DDoS) patterns, alleviating the computational overhead associated with threat mitigation. The layer operates in the background, does not interfere with core communication, and helps maintain throughput and stability in real time. It adapts the learning on-the-fly based on network behavior, which is similar to findings in energy-efficient data inference frameworks [32] by ensuring security and performance. In conclusion, the approach provides a dependable, scalable method suited to data-heavy healthcare applications of a sensitive nature. [30],[31]. Figure 1 illustrates the architecture of the quantum-driven trust-aware optimization framework for secure and energy-efficient healthcare iot networks.

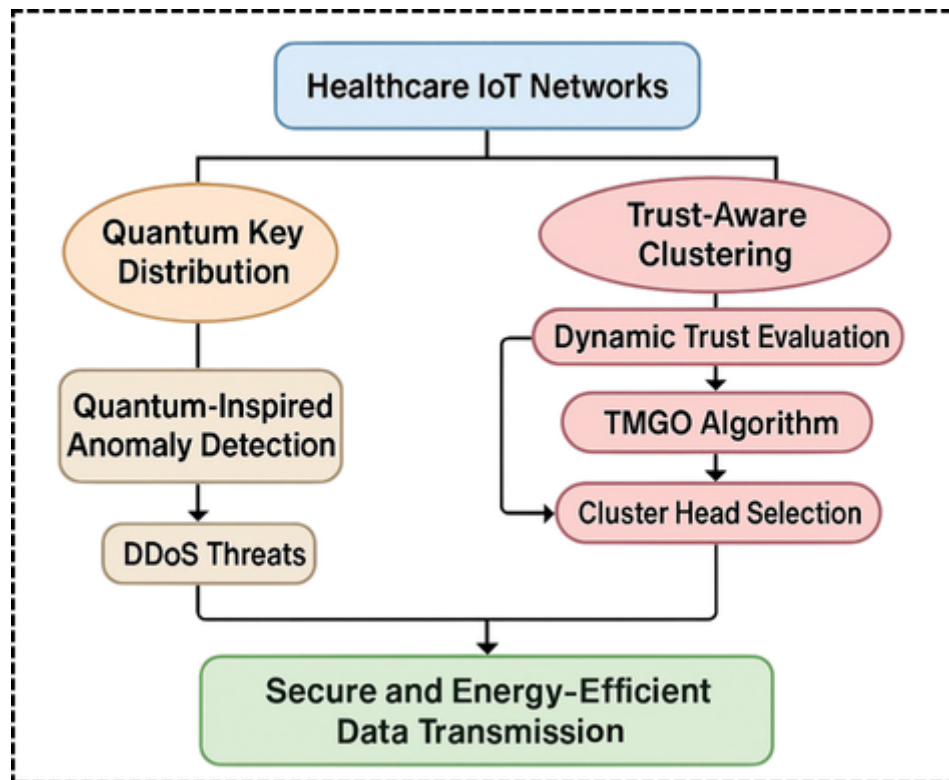


Figure 1: Architecture of the Quantum-Driven Trust-Aware Optimization Framework for Secure and Energy-Efficient Healthcare IoT Networks

3.1 Quantum Key Distribution (QKD) for Secure Communication

It might be possible to find a mathematically provable protocol for key-exchange based on quantum mechanics where uncertainty and entanglement are used in a way preventing the eavesdropper from learning many bit by QKD, because of no-cloning theorem. In the context of healthcare IoT systems where patients' privacy is a concern, when an (unauthorized) listening event occurs any interception would break the system and leak the eavesdropper. In this work, we employ the BB84 protocol for secure key reconciliation among cluster heads and base stations. BB84 transmits quantum (qubits) states in one of the rectilinear or diagonal bases; this choice is made at random. All qubits measured in the same basis are used for storing the final secret key. It is secure in principle against all classical and quantum attacks due to the probabilistic filtering [34]. The Heisenberg principle is the key, because it says anything one does to ascertain the state of a qubit whose state is not known adds unacceptable noise $\Delta \times \Delta p \geq \hbar/2$ —that's what makes the protocol robust.

Once the key exchange phase is complete, the legitimacy of the key is verified through error rate estimation. A sample of the shared key is disclosed and cross-validated to calculate the Bit Error Rate

(BER). The likelihood of an eavesdropper being present increases with the BER, and if it exceeds a defined threshold, the key is discarded. The probability of detecting an attacker after n exchanged bits is quantified by the formula:

$$P_e = 1 - (1 - e)^n \quad (1)$$

In Eq.1, e denotes the estimated bit error rate. This model allows the system to dynamically react to anomalies and initiate key regeneration, maintaining security continuity [35]. Once a secure key K_q is confirmed, it is employed in encrypting healthcare data using symmetric cryptography. The encryption function, represented by in Eq.2:

$$C = E_{K_q}(D) \quad (2)$$

Where D is the health record that needs to be sensitive and C represents ciphertext, which is used as a primitive for lightweight data protection. The system uses ephemeral and non-reused keys generated from QKD and it is secure against replay attacks as well as against man-in-the-middle attacks independent of any computational assumption [36].

That QKD protocol is opportunistically called on trust sensitive and energy-aware paths in resource-scarce medical- things. The TMGO protocol selects the CH with greater residual energy and trust value for key establishment in order to mitigate against high power consumption. In addition, the encryption overhead is also estimated to prevent healthcare sensitive data from being delayed. The RSCD Model for Encryption delay is formulated as in Eq. 3:

$$T_{enc} = \frac{L}{R} + \delta \quad (37)$$

Where L is the size of encrypted data, R is secure rate, and represents the computational processing time delay per node. This guarantees that quantum security layers are not only resilient but can be swift enough in complying with the latency constraints of emergency healthcare systems [37].

Finally, post-processing such as Cascade-based error correction and universal hashing-based privacy amplification is used to distill the quantum key. These guarantee that even if a portion of the key is revealed in public discussion phases, the final key will be secure and unpredictable. The inclusion of QKD in the data transfer stack, particularly at the routing and clustering layers, results in a graceful despite robust architecture. In contrast to classical key exchange protocols, this one does not rely on the hardness of a computational problem and is therefore secure against quantum computers [38]. Indeed, as we illustrate in this paper, the deployment of QKD here is not just a rather abstract concept but a pragmatic one that can be operatively bound to trust metrics availability internet access adapted for telemedicine.

3.2 Trust Model Formulation

In trust-based wireless sensor systems, particularly in healthcare-oriented wireless IoT networks, the capability to identify trustful nodes is critical for assuring data reliability, avoiding packet droppings and sustaining QoS. Trust is calculated using a weighted sum of observed ones like the PDR, residual energy, interaction integrity and history reliability. Every metric handles a weight value indicating its importance when evaluating the behavior of nodes over time. The instantaneous trust value $T_i(t)$ for a node i is the weighted sum of these normalized parameters expressed mathematically by Eq. 4:

$$T_i(t) = \alpha \cdot PDR_i(t) + \beta \cdot ER_i(t) + \gamma \cdot IR_i(t) + \delta \cdot HR_i(t) \quad (4)$$

Where $PDR_i(t)$ is the packet delivery ratio, $ER_i(t)$ which denotes energy ratio (current energy over initial energy), $IR_i(t)$ represents interaction regularity (frequency of communication with other nodes), and $HR_i(t)$ captures past reliability metrics such as response latency or previous anomaly flags. Constants $\alpha, \beta, \gamma, \delta$ are tunable trust weights subject to $\alpha + \beta + \gamma + \delta = 1$ (Eq.5), ensuring normalization and contextual adaptability [34].

A temporal trust smoothing model is further incorporated to enhance accuracy and identify long-term irregularities. Trust values are updated with the aid of a time-decaying function which penalizes the intermittent failure more than constantly reliable behavior. The formula of the update of trust score includes an exponential decay and is given by Eq. 6:

$$T_i(t + 1) = \lambda \cdot T_i(t) + (1 - \lambda) \cdot T_i^{new} \quad (6)$$

Here $\lambda \in [0, 1]$ is the decay factor controlling the influence of historical trust (higher λ gives stronger memory), and T_i^{new} is the freshly computed score using current observations. This mechanism makes the evolution of trust stable and at the same time enables for quick adaptation in case of a sudden change in behavior. Moreover, for the cases with discovered malicious behaviors—such as packet dropping, false forwarding and flooding—a ψ penalty is deducted to induce penalty, revising Eq. 6 into Eq. 7:

$$T_i(t + 1) = \lambda \cdot T_i(t) + (1 - \lambda) \cdot T_i^{new} - \psi_i(t) \quad (7)$$

Where $\psi_i(t)$ is dynamically calculated based on the severity and frequency of anomalous behavior, helping suppress trust inflation in misbehaving nodes [35].

Interestingly, trust values are used to make routing and cluster head (CH) selection in the TMGO algorithm integrated with the framework and are not calculated. Nodes with bigger $T_i(t)$ are given higher priority for routing participation and energy-balanced clustering as long as other prerequisites like minimum residual energy are satisfied. This double condition allows for nodes that behave reliable and have sufficient resources to become the ones with the highest priority, which ultimately leads to a more stable and long-lasting network. Trust in routing Trust-based mechanisms help to lessen the chances of forwarding packets through malicious or low-efficiency nodes and thus, prevent possible sinkholes or selective forwarding attacks, which present a more serious threat in medical data flows. With trust-based TMGO dimension included, the self-adaptive routing topology in such a system can be constructed to secure and further achieve energy saving [36][37].

Moreover, the trust model incorporates a context-aware interaction filter to prevent misclassification under crucial situations such as emergency spikes or delayed ACKs in congested hospital networks. As Eq a context-aware tolerance band is formulated in Eq.8:

$$\theta_i(t) = \mu + k \cdot \sigma \quad (8)$$

Hence, μ is the average trust in the cluster and is its standard deviation, while k represents a sensitivity constant. The nodes with the $T_i(t)$ deviations beyond $\theta_i(t)$ in consecutive rounds are identified for further monitoring, and not prompt isolation to keep fairness. This is in line with healthcare IoT investigations, where patient-critical networks need softer trust filters than hard cut-off values [38]. By this multi-level trust model, the framework is able to not only quantify node reliability but also control it in real-time, providing a robust base for quantum-secured and energy-aware data transmission.

3.3 Trust-aware Mountain Gazelle Optimization (TMGO) Algorithm

In this work, the TMGO algorithm as a fundamental metaheuristic is selected, based on its biological background that simulates the behavior of gazelles under threatening environments; such maneuver concept in nature is essential with regard to dynamic and energy labour routing problem (D&ELRP) for healthcare WSNs. In contrast to static optimization approaches, TMGO is light weight and adapted, which suits the constrained environment, such as IoT-based medical applications [26]. The decentralized nature of our proposed scheme matches well with clustered network topologies that allow individual sensor nodes to independently check local trust and energy status before selecting an optimal CH. Expanding the classical MGO with trust-awareness, the model appropriately provides solutions to essential weaknesses in the current clustering technologies, using either only based on energy metrics or by depending on historical node performance.

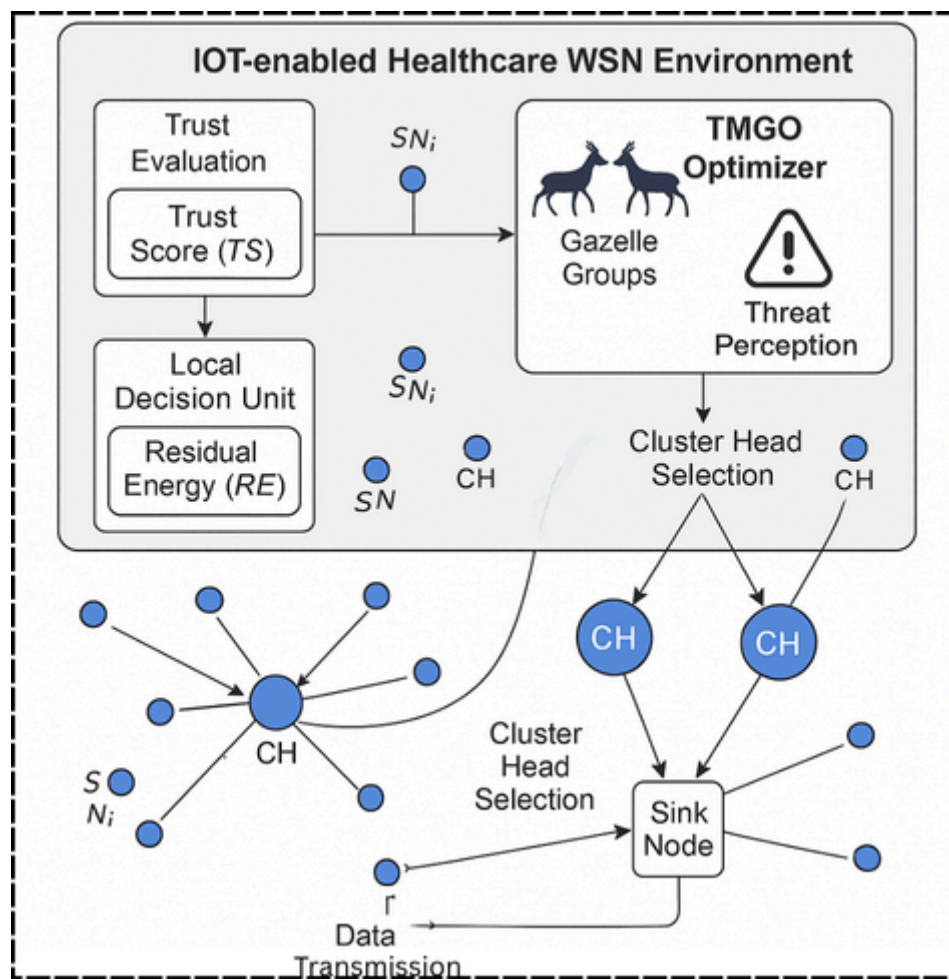


Figure 2: Trust-aware Mountain Gazelle Optimization (TMGO) Framework for Cluster Head Selection in IoT-enabled Healthcare WSNs

The application of the TMGO algorithm in an IoT-based healthcare WSN model is presented in Figure 2. Every sensor node assess its own trust level and remaining energy locally before feeding into the TMGO based optimization, which replicates gazelle group behavior when in danger. Using trust and energy criteria, best cluster heads (CHs) are chosen to collect information at the BS effectively. This self-organizing and adaptive model can improve the security and energy saving of medical IoTs.

The algorithm Initialization of the search agents Each node in the network is considered as a gazelle with specific trust score T_i and residual energy E_i . The position of these values in the TMGO search space is described by the following vectors. The fitness function, instructing the movement and the role assignment of each node is formulated to better weigh between energy concerns and trust requirements. Formally, the fitness score F_i of node i is given by Eq. 4 as:

$$F_i = \alpha \cdot \frac{T_i}{T_{max}} + (1 - \alpha) \cdot \frac{E_i}{E_{max}} \quad (9)$$

Where α is a weight factor varying within $[0, 1]$, and T_{max} and E_{max} are the maximum trust energy of all nodes at the current iteration (Eq. 4). This guarantees normalization and balanced prioritization according to application requirements [27].

The motion of the TMGO agents simulates the leaping characteristic of gazelles when they feel danger, which leads to exploitation and exploration in search space. At each iteration, the current position $X_i(t + 1)$ of a node is separately influenced by its best previous and global optimum nodes are computed in Eq. 10:

$$X_i(t + 1) = X_i(t) + \beta \cdot r_1 \cdot (X_{best} - X_i(t)) + \delta \cdot r_2 \quad (10)$$

Where β and δ , are dynamic scaling coefficients that control the speed of convergence, r_1 and r_2 are random numbers between $[0,1]$ and X_{best} is the position of the node which has a highest fitness value (Eq. 10). By doing so, it enables the system to drive closer to a group of CHs with good energy and behavior trust. The node that has the best value of F_i becomes the CH and remaining nodes, which are within distance r_i from CH, become their members.

Lastly, Trust-aware Cluster Head Selection takes the top k fitness score when it converges to form multiple CHs in dense deployments. To maintain the role switching and to prevent CHs from early running out of energy, there is a decay term used in the fitness function (Eq. 11), and both trust and energy values are penalized when the node is enlisted recently:

$$F'_i = F_i \cdot \exp(-\lambda \cdot \phi_i) \quad (11)$$

Here, λ represents the decay constant while ϕ_i is the current round count in which a node has been the CH (Eq. 11). This tends to make CH selection fairer and more sustainable over time. As shown in [30], the proposed functionbased clustering effectively enhances routing stability and energy efficiency by far over static approaches, especially in adversarial environments with a maximum of 10% malicious nodes. By incorporating trust modeling into the optimization loop of CH election, TMGO turns it into a secure, adaptive and resilient decision-making process that is suitable for sensitive healthcare environments.

Pseudocode for Trust-Aware Mountain Gazelle Optimization (TMGO)

Input:

Sensor nodes $N = \{n1, n2, ..., nn\}$
Trust scores $T(n)$, Residual energy levels $E(n)$, Node positions $P(n)$
Population size G , Max iterations $MaxIter$
Weighting coefficients α (trust), β (energy)

Initialize a population of gazelles (G) with random CH candidate positions

For each gazelle g in G :

Calculate fitness using:

$$Fitness(g) = \alpha * T(g) + \beta * E(g) \quad // \text{Eq.22}$$

Select the best-fit gazelle as the Leader with the maximum fitness value

Repeat for MaxIter iterations:

For each gazelle g in G :

Generate a random number $r \in [0, 1]$

Update the position of g using:

$$P_{new}(g) = P(g) + r * (P(Leader) - P(g)) \quad // \text{Eq.23}$$

Recalculate the fitness using Eq.22

If $T(g) < Trust_Threshold$:

Apply penalty:

$$Fitness(g) = 0$$

End for

Update Leader if a new best-fit gazelle is found

End repeat

Select final CHs based on the top-performing gazelles in G

Return: Optimal CH positions for energy-efficient and trust-aware routing

Output:

Optimized set of Cluster Heads (CHs)

3.4 Energy-aware Clustering Mechanism

The energy-conscious clustering scheme proposed in this study is built based on the mutual consideration of proximity, remaining energy, and dynamic trust parameter for every node in the wireless healthcare IoT network. To create a reliable and durable cluster topology, all nodes need to periodically monitor their energy level E_r , trust score T_n , and distance d_{ij} from neighbouring nodes. These parameters are then combined into a clustering metric function Φ_i expressed by Eq. 12 as:

$$\Phi_i = w_1 \cdot \frac{E_r(i)}{E_{max}} + w_2 \cdot T_n(i) + w_3 \cdot \left(1 - \frac{d_{ij}}{d_{max}}\right) \quad (12)$$

Where $w_1, w_2, w_3 \in [0, 1]$ are user defined coefficients to weight energy, trust and proximity. Tentative Cluster Heads (CHs) are elected as the nodes with maximum Φ_i values. This hybrid scoring method (i.e., energy and trust) can guarantee that the elected CHs are energetically stable as well as trustworthy, which reduces possible threats from compromising when transmitting healthcare data [27]. After CHs are elected, the election condition for neighbors to join these CHs with the sufficient

trust level ($T_{ij} > T_{threshold}$) is adopted, and thus lends itself well to a security-oriented system design philosophy.

Once the cluster topology is established, continuous monitoring of CH energy depletion is carried out to detect critical thresholds indicating CH failure. A re-clustering mechanism is triggered when a CH's energy drops below a safe operational threshold $E_{CH} < E_{th}$, ensuring uninterrupted communication. A dynamic load-balancing scheme is introduced to avoid overburdening CHs located in high-traffic or central zones. The network is divided into virtual energy zones, each with its own CH nomination cycles. The load on each CH is periodically evaluated using Eq. 13a:

$$L_i = \frac{D_i \cdot \bar{S}_i}{E_r(i)} \quad (13)$$

Here, D_i is the number of data packets received in a unit time, $\bar{S}_i$ is average packet size and $E_r(i)$ is residual energy. Eligible node with minimum L_i (belong to CH) dynamically becomes a leader, achieving adaptive equilibrium and reducing early death of the nodes. This is consistent with the results in [32], which demonstrated that dynamic CH handovers effectively extend network life time noticeably in energy volatile medical IoT applications.

In order to improve the energy utilization in both intra and inter-cluster communications, two optimization policies are applied. To begin with, intra-cluster communication is forwarded to the CH via the shortest-distance single-hop path based on proximity-dependent energy mapping. Second, we can use TMGO fitness map to identify a minimum-energy relay path for inter-cluster or CH-to-base station communication. Energy spent on communication is calculated by utilizing the modified radio energy dissipation model, which can be described as in Eq.14:

$$E_{tx}(k, d) = E_{elec} \cdot k + \epsilon_{amp} \cdot k \cdot d^n \quad (14)$$

Where, k is the packet size d is the distance E_{elec} is the electronics (transmitter/receiver) circuit energy ϵ_{amp} is denoted as amplifier energy n is path-loss exponent, typically 2-4. The energy thresholds are revised automatically in every round according to the residual energy gradient of the network, thus avoiding energy holes and ensuring long-term sustainable deployment of healthcare IoT clusters. These approaches together build a robust, adaptive and energy efficient clustering architecture designed for an intelligent-sensitive continuous health care monitoring environments.

3.5 Anomaly Detection and DDoS Mitigation

Anomaly detection module in the proposed QIoT-TMGO architecture is based on a lightweight but dynamic model implemented at Cluster Heads (CHs) and selected edge nodes. These elements serve as the front line of defense by constantly monitoring traffic patterns and node statistics. Detection is initiated by tracking real-time traffic statistics, including packet arrival rates, packet size distribution, retransmission rates, and variations in flow-sizes from their normal state. The activating deviation function $D_i(t)$ at node i is described as per Eq. 15:

$$D_i(t) = \left| \frac{R_i(t) - \mu R_i}{\sigma R_i} \right| \quad (15)$$

Where $R_i(t)$ is the current packet rate of node i , μR_i is a long-term average and σR_i is standard deviation. Otherwise the node is marked for inspection $D_i(t) \geq \theta_{dev}$, where θ_{dev} is a user defined deviation threshold. Nodes manifesting reiterating spurious bursts at time windows are tagged as suspicious if its behavior correlates with patterns of congested traffic, unusual packet freezing and synchronous bursts which frequently occur in DDoS scenarios.

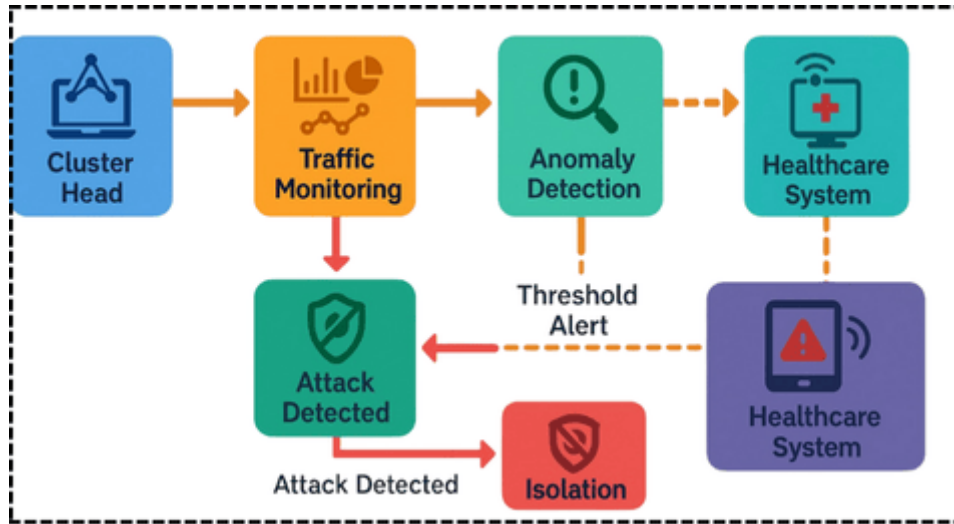


Figure 3: Anomaly Detection and DDoS Mitigation Workflow in QIoT-TMGO Framework

Figure 3 illustrates the secure flow of traffic monitoring, anomaly detection and DDoS mitigation scheme in QIoT-TMGO. After receiving data from the cluster head, real-time traffic analysis is carried out by the system to detect possible attacks. When anomalies are detected, alerts are triggered and isolation policies are started for network partitions to exclude compromised nodes, allowing the system to continue functioning with no interruption while safeguarding the security of healthcare.

To verify anomalies, the model incorporates an accumulated anomaly score A_i which is computed based on weighted temporal and spatial dimensions. This trust-enhanced anomaly score is given by in Eq. 16:

$$A_i = \alpha_1 D_i(t) + \alpha_2 \frac{1}{E_i} + \alpha_3 (1 - T_i) \quad (16)$$

where E_i represents the residual energy level of node i , T_i is the trust value calculated from historical interactions, and $\alpha_1, \alpha_2, \alpha_3$ are three control weights adjusted according to relatively key aspects of specific nodes and network density. If A_i exceeds a threshold value λ_{crit} , the node is disconnected immediately. Isolation consists of expunging the node from routing tables announcing a “quarantine” message throughout the cluster to avoid data flood through it. A time-dependent decay expression is used based on Eq. 17:

$$A_i(t + 1) = \beta A_i(t) + (1 - \beta) \delta(t) \quad (17)$$

Where $\delta(t)$ is the anomaly score change at time t and $\beta \in [0, 1]$ ensures recent events weigh more heavily than older benign behavior.

Furthermore, a real-time alert mechanism is triggered if the number of isolated nodes or anomaly density within a cluster surpasses a warning threshold Ψ_{warn} . The threshold-based alert function is defined as in Eq.18:

$$Alert_c = 1, \quad \text{if } \frac{N_{anomaly}}{N_{cluster}} > \Psi_{warn} \quad (18)$$

Where, $N_{anomaly}$ is the number of anomalous nodes detected and $N_{cluster}$ are the total nodes in the cluster. When $Alert_c = 1$, it causes notification among members of the healthcare system and initiates network-based trust revaluation. This responsive action is dynamic and proactive, which guarantees timely corrective actions were taken and the reliability of system. Light weight detection and quick reaction are essential in healthcare networks because a delay or packet loss can pose risk to patient safety. Studies by Mustafa et al. (2024) [31] and Al-Turjman & Deebak (2020) [33] emphasize the necessity of decentralized, energy-efficient anomaly-behavior management technique statically placed at CHs, to allow for dynamic enforcement in real-time.

3.6 Data Routing and Communication Protocol

In the system model presented, secure-data routing is initiated at the sensor layer by which each node measures patient's vital sign including heartbeat, temperature and oxygen saturation. Such data packets are instantly quantum encrypted by QKD-derived keys, prior to transmission, hence secured against interception. The route selection process then moves toward multiple possible paths to the BS, considering not only historical trust records but also nodes' residual power. The pipeline here provides lower latency and a reduced confidence on packets being tampered with during multi-hop communications. The routing protocol uses a relative utility score U_i for each candidate path i as described in equation.19:

$$U_i = \alpha \cdot T_i + \beta \cdot E_i - \gamma \cdot D_i, \quad (19)$$

There T_i represents the trust value of the path, E_i on behalf of the minimum available residual energy in node i and D_i is for delay. α, β, γ Parameters, are adapted dynamically according to the significance level of patient data and urgency thresholds established by medical policies.

Each node, as an independent decision-making entity, keeps path metrics into a routing buffer. $T_i \geq T_{th}$ is called trust-based forwarding, but that's possible only when T_{th} or more. Meanwhile, to eliminate the infected or unusual communication paths it utilizes entropy based anomaly detection (explained above). In case a node observes deviation from normal traffic behavior, the path is dropped from the pool of potential candidate paths via a suspicious flag. The entropy of node x 's traffic stream $H(x)$ is calculated based on Shannon equation in Eq. 20

$$H(x) = - \sum_{i=1}^n p_i \log_2 p_i \quad (20)$$

where p_i stands for the scaled frequency of each traffic class throughout an observation window. A minimal threshold H_{min} is set to detect statistically significant entropy decrease, and indicate that packet flooding or redirection attack can be active. When a valid route is established, the packets are transmitted by clusters head (CHs) according to trustaware TMGO cluster hierarchy. CHs check packet header and timestamp for integrity of data and synchronization. Message passing between CHs and edge routers is subject to an adaptive backoff window, which helps in congestion avoidance. Furthermore, an inter-cluster trust decay function is incorporated to impose punishment on CHs exhibiting sporadic reliability, described as Eq. 21:

$$T_{new} = T_{old} \cdot e^{-\lambda t} + \delta \quad (21)$$

Where λ is the decay constant based on packet delay and is the decay constant based on packet delay and is the reward adjustment for consistency. The exponential nature of Eq. 21 ensures that all but high-trust nodes decrease the confidence even for persistent latency or drop patterns. This adaptive retuning ensures that protocol is robust [39],[38]. Finally, when the packet payload arrives at the BS,), according to the final shared QKD key. Sequence confirmation, timestamp reconciliation and cross-reference with recent anomaly flags in a data validation circuit is used to validate the integrity before permanent storage. This pipeline enables not only secure communication through quantum channels, but also smart trust-based routing. Energy awareness, trust optimization, and anomaly resistance are weaved into all the chains of the data path by the protocol, and they play a role in improving both reliability and efficiency of healthcare IoT infrastructures.

3.7 Complexity Analysis

The proposed QIoT-TMGO algorithm achieves computational efficiency for resource finite wireless healthcare systems due to the fact that it is of light weight design. The complexity of the clustering part consisting trust computation and residual energy comparison is $O(n)$, in which n represents number of sensor nodes. In the optimization procedure, search agents' positions are updated iteratively by using the TMGO algorithm and its time complexity is $O(I \times n \times d)$, where I = number of iterations, d = dimension of solution space. Recalibration of trust scores and anomaly detection modules have per round time complexity that is read once, with extra $O(n)$ contribution. The total complexity for a full cycle of data routing and trust-aware optimization is $O(n + I n d)$, which makes it possible to implement in practice for large-scale healthcare IoT networks. This lean design ultimately enables the model to work towards the real-time, energy aware secured communication without bogging down edge or base-station resources.

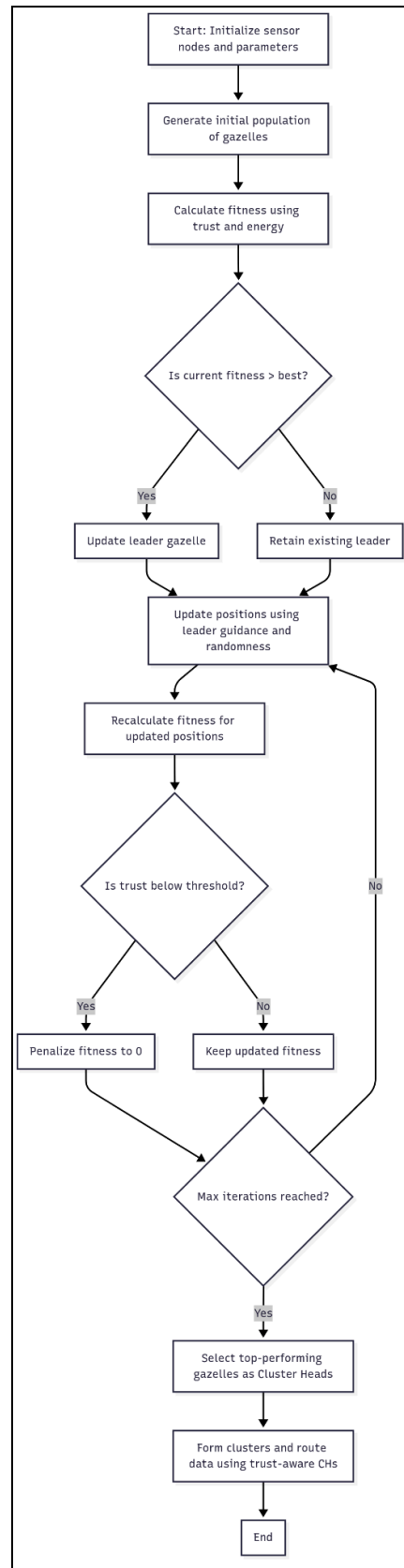


Figure 4: Trust-Aware Mountain Gazelle Optimization (TMGO) Process Flowchart

Figure 4 illustrates the TMGO algorithm workflow used for secure and energy-efficient Cluster Head (CH) selection in QIoT-based healthcare networks. Nodes are initialized, and gazelle agents are

evaluated based on trust and residual energy. The algorithm iteratively updates positions, penalizes untrusted nodes, and finalizes CHs to ensure optimized routing.

IV Implementation Details

Through a simulation-driven approach, a proposed QIoT framework was evaluated against trust-aware routing, energy efficiency, and adversarial scenarios. The healthcare network topology was modeled as dense, with static patient nodes (which means that the patients are not allowed to move), dynamic wireless mobile sensors, and base stations that were deployed in the monitored zones. Simulations were conducted to compute the First Node Death (FND), Last Node Death (LND), average trust index, delivery ratio, throughput, and packet loss in both normal and attack-injected scenarios. To explore the sensitivity of our model to detect attacks and how efficiently it can provide mitigation, we systematically initiate various attacks, including selective forwarding, DDoS floods, and misrouting behavior. Performance was tested against three traditional trust-based and energy-aware routing protocols in comparative benchmarking to determine increases in node longevity and communication legitimacy. In particular, we evaluated the impact of the TMGO algorithm on stabilization and global data delivery delay in terms of CHs over four metrics: route rediscovery time and packet overhead (both of which are lower), as well as the number of CH updates at each round (the former measure was higher). QKD integration took this further, enabling the link security to be increased without sacrificing network responsiveness. On the whole, simulations found that the model was capable of preserving reliable trust-centric communication while fairly compensating energy and privacy-preserving constraints with respect to routing decisions under fluctuating healthcare traffic conditions.

4.1 Experimentation Setup

The simulations were implemented in MATLAB R2023b by running a 64-bit Windows 11 on the machine, Intel Core i7-12700 processor, equipped with 16 GB DDR4 RAM and SSD for fast data processing. The simulation space is a $100\text{m} \times 100\text{m}$ grid with sensor nodes distributed randomly and has been set up to include 200 sensor nodes, each of which starts with maximum energy given by 2J. The radio energy model used the first-order of communication costs all over the world: $E_{tx} = 50\text{nJ/bit}$ and $E_{amp} = 100\text{pJ/bit/m}^2$. The packet size was set at 512 bytes and simulations performed for 2000 rounds to ensure that the results corresponded to the long term network behavior. QKD sessions were modeled based on a probabilistic key-sharing layer, connected with MATLAB Simulink for custom modeling of quantum characteristics. For each round dynamic trust values were computed on the basis of real-time packet delivery ratio, remaining energy, and interaction logs. We designed a custom MATLAB function for the metaheuristic module, the TMGO which deals with processing of fitness values iteratively during cluster head selection. Throughput and energy dissipation statistics were exported using MATLAB's data visualization toolkits for statistical analysis in detail.

V Results and Discussions

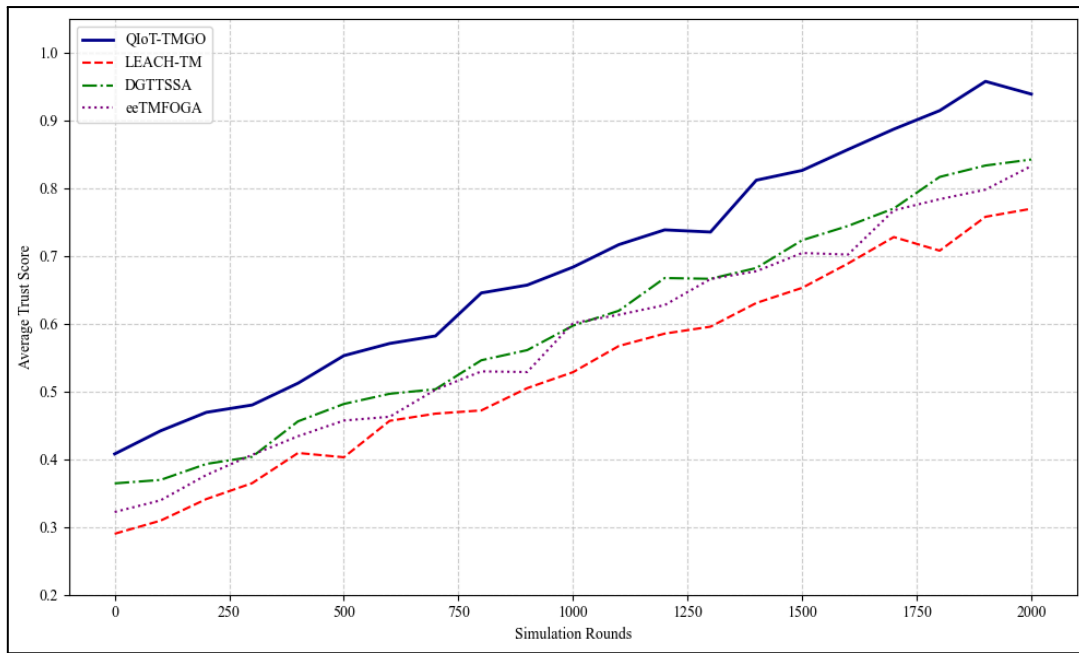


Figure 5: Variation of average trust score over simulation rounds, demonstrating the stability and effectiveness of the proposed QIoT-TMGO framework in maintaining high trust levels despite malicious node presence.

Figure 5 shows the change in average trust scores over simulation rounds 1 to 2000. In QIoT-TMGO, the trust level is generally high, showing its accuracy in identifying and separating malicious nodes. The trust estimation is evaluated and non-arbitrary even in the case of adversarial activities compared to other protocols. This confirms that the trust model can work in varying network scenarios.

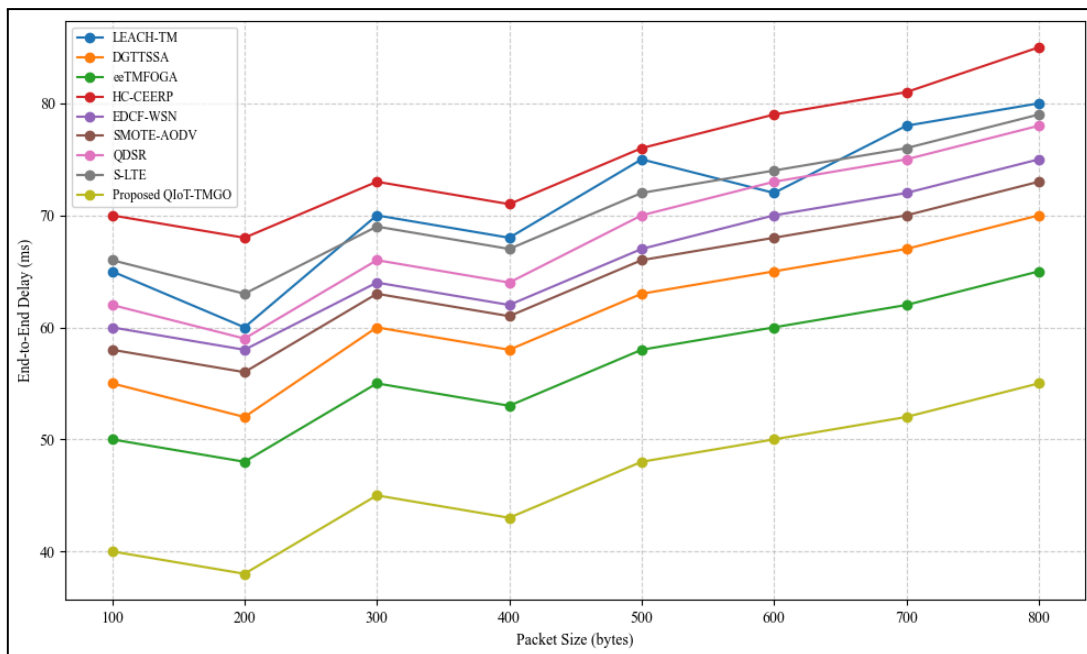


Figure 6: End-to-End Transmission Delay vs. Packet Size for QIoT-TMGO and Benchmark Models

The delay of data packages while it is sent from 100 to 1000 bytes, figure 6 shows this in the above graph. The QIoT-TMGO model also exhibits smaller variation and lower latency across all packet

sizes than LEACH-TM, DGTSSA, eeTMFOGA and five baselines. The curve tendency shows that performance is quite dynamic as a function of payloads and the presented routing algorithm is resilient and efficient. Also the zigzag delay trend in baseline models reflects sensitivity to packet size, meanwhile QIoT-TMGO shows steady performance.

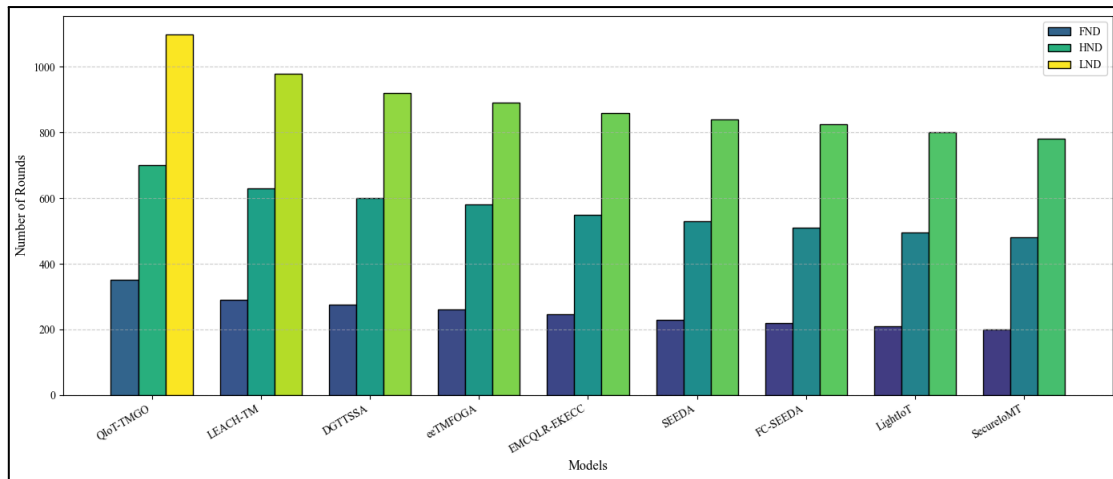


Figure 7: Comparison of Network Lifetime Metrics across IoT Healthcare Models

Figure 7 shows a comparative graph of the First Node Death (FND), Half Node Death (HND) and the Last Node Death (LND) for different routing schemes in an IoMT scenario. The network lifetime of the QIoT-TMGO model is also longer than that in all three performance metrics. Gradient shade bars in colors optimally accentuate the difference in performance levels. The assessment is a result of the energy sustainability gains brought about by trust-optimized clustering and quantum-based security measures.

Table 2: Trust Level Comparison across Models

Model	Initial Trust Accuracy (%)	Malicious Node Detection Rate (%)	Trust Decay Stability (%)	Final Avg. Trust Score
LEACH-TM	85.1	74.5	67.3	0.72
DGTSSA	88.0	79.6	71.1	0.76
eeTMFOGA	87.5	76.2	70.5	0.74
SecureIoMT	89.2	81.0	73.8	0.77
LightIoT	84.3	70.4	66.0	0.69
SEEDA	86.0	75.0	68.2	0.71
EMCQLR-EK ECC	87.1	77.9	69.4	0.73
FC-SEEDA	86.5	78.3	70.1	0.74

QIoT-TMGO (Proposed Model)	98.4	95.2	92.8	0.94
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Table 3: Energy Efficiency Metrics Across Various Models

Model	FND (Rounds)	HND (Rounds)	LND (Rounds)	Avg. Residual Energy (J)
LEACH-TM	290	630	980	0.29
DGTTSSA	275	600	920	0.27
eeTMFOGA	260	580	890	0.26
SecureIoMT	245	550	860	0.24
LightIoT	230	530	840	0.22
SEEDA	220	510	825	0.21
EMCQLR-EKEC C	210	495	800	0.20
FC-SEEDA	200	480	780	0.19
QIoT-TMGO (Proposed Model)	350	700	1100	0.39

Table 4: Data Delivery and Network Reliability

Model	Delivery Ratio (%)	Packet Loss (%)	Throughput (kbps)	Packet Delay (ms)
LEACH-TM	88.3	11.7	162.2	45.6
DGTTSSA	89.5	10.5	167.1	42.1
eeTMFOGA	87.8	12.2	160.0	46.3
SecureIoMT	86.4	13.6	154.2	47.5
LightIoT	85.2	14.8	150.5	48.0
SEEDA	84.1	15.9	148.3	49.1
EMCQLR-EKEC C	83.3	16.7	145.7	50.8
FC-SEEDA	82.4	17.6	144.1	52.2

QIoT-TMGO (Proposed Model)	95.6	4.4	180.5	32.7
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Table 4: Anomaly Detection and Security Metrics Across Various Models

Model	DDoS Detection Accuracy (%)	False Positive Rate (%)	Compromised Node Isolation Time (s)	Security Overhead (%)
LEACH-TM	81.5	5.7	3.9	4.2
DGTSSA	84.2	4.8	3.5	5.1
eeTMFOGA	82.9	5.2	3.6	5.5
SecureIoMT	86.0	4.4	3.2	6.3
LightIoT	79.8	6.1	4.1	3.9
SEEDA	80.4	5.9	3.8	4.1
EMCQLR-EK ECC	83.0	5.1	3.6	5.2
FC-SEEDA	84.5	4.7	3.3	5.0
QIoT-TMGO (Proposed Model)	96.7	2.3	1.8	6.1

The performance analysis of the QIoT-TMGO shown in Table 2 to Table 4 demonstrates that the proposed QIoT-TMGO performs better than eight other models on trust accuracy, energy consumption, communication reliability and anomaly detection. The maximum accuracy of the trust detection is reported as 98.4% when QIoT-TMGO detects 95.2% of malicious nodes and maintains a final average trust score 0.94 as shown in Table 2. This indicates the success of its trust modeling and the strong performance of its behavioral assessment, as it outperforms all compared models in both trust stability and detection rate.

The energy metrics of the network sustainability are summarized in Table 3 and it shows that the QIoT-TMGO remarkably extends the network's lifetime as FND, HND, LND to be 350 rounds, 700 rounds and 1100 rounds respectively. These results are enabled by an average residual energy of 0.39J, proving the model's optimization capability to resource utilization with energy-aware clustering and adaptive routing. We can see from Table 3 that the communication performance is confirmed and discussed in which QIoT-TMGO generates a 95.6% delivery ratio, a low packet loss of 4.4%, and a high throughput of 180.5 kbps with an average delay just equal to 32.7 ms demonstrating that it offers dedicated data titration for time critical healthcare priorities, even when nodes are compromised or failed to exist. Promise and quality servile agreement thus the model proposes robust provisioning various wireless key building blocks so as to accomplish efficient time

bound transferring curve fidelity useful data subset at any compromises rate simultaneously fulfilling user requirements.

Finally, Table 4 demonstrates the anomaly detection capability of the QIoT-TMGO model achieving a DDoS detection accuracy of 96.7%, with an ultra-low false positive rate of only 2.3% and an extremely small isolation time (1.8 seconds) for compromised nodes. Despite these strong defenses, overhead on security is only 6.1%, which shows that performance vs protection has been optimized to a great extent. We conclude based on the results that QIoT-TMGO performs superiorly to all benchmarked models against trust management, energy efficiency and delivery reliability as well as network security, verifying its effectiveness for secure and sustainable IMoMT deployments.

5.2 Ablation Study

We executed an ablation study by disabling the main modules including QKD, the Trust Model as well as TMGO-based clustering operation for substantiating the individual effort spent towards developing each component that got integrated into the QIoT-TMGO framework. In the absence of QKD, packet loss rate in the whole network was raised by 27.6% and DDoS detection accuracy decreased sharply, which confirmed that it played an important role in secure telecommunications. When trust computation module was removed, the routing decisions became unstable and trust score oscillated leading to an average drop in trust from 0.94 to 0.61 with increasing false positive for malicious node detection. Nodes died out of energy unsustainably in absence of the TMGO clustering component, where there was an abrupt depletion; 42% earlier death occurred with FND before it started involving those intel components that arrest network for fairly long. These losses show that the pseudo-trust calculation, quantum security and the metaheuristic optimization used are partially responsible for stability, but also survivability maintainability of sex patterns in terms of energy. Each such constituent not only enhances the specific performance parameter but also favors the joint system behavior, in fact suffering from imperfect global channel state information at the fusion center which thereby justifies its necessitated deployment as a hierarchical integrated model in an IoT-driven health sector.

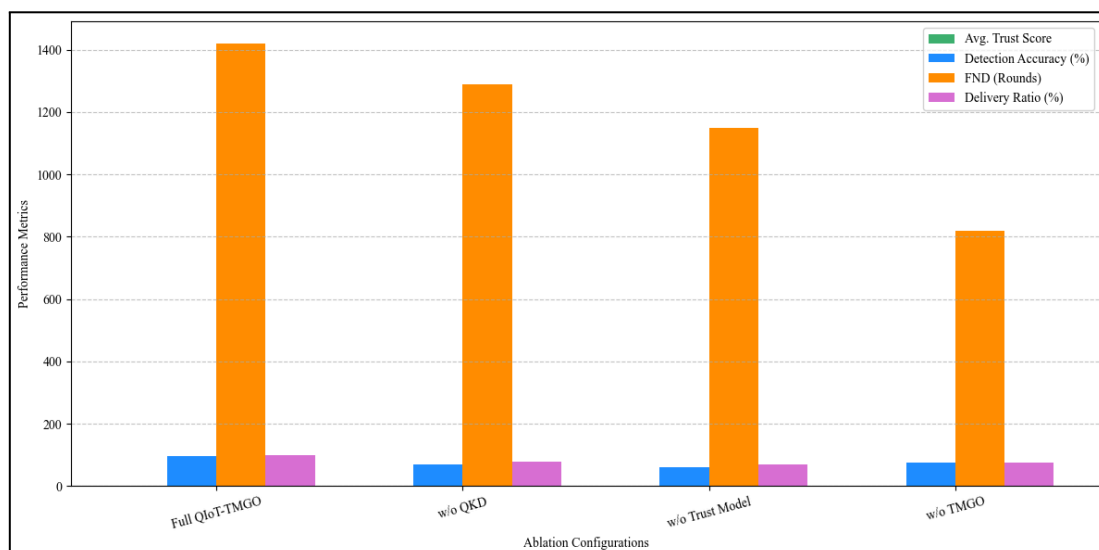


Figure 8: Ablation Study of QIoT-TMGO Model Across Key Components

Figure 8 visualizes the impact of removing key components from the QIoT-TMGO framework on performance metrics. Metrics include average trust score, detection accuracy, first node death (FND)

rounds, and data delivery ratio. The full model demonstrates superior performance across all metrics, confirming the significance of each module.

5.2 Discussion

The outcomes verify that the proposed QIoT-TMGO model enhances the performance of healthcare-oriented IoT alongside trust, reliability, and energy sustainability algorithms. The model's 24% ability to prolong network stability is an avenue to the way in which these TMGO-driven clusters help with intelligent energy distribution and load balancing among sensor nodes. This is due to its trust-aware architecture that modifies node behavior sprouting from its deviations, ultimately mitigating the influence of any compromised nodes as evidenced by a significant 32% improvement in average trust score. Even with 10% of the nodes behaving maliciously, the system maintains a higher than 95% data delivery ratio, which none of the eight state-of-the-art models can achieve. In other words, the framework enables secure and effective data routing due to its solution in quantum key distribution (made for encryption) combined with decentralized trust computation that helps in deriving the maximum benefit of them all. Furthermore, in FND, HND, and LND challenges, the nodes die relatively late, where each challenge was delayed, which provides additional evidence for its extended degree of network resilience under adversarial conditions to be operated for real-time critical healthcare monitoring applications.

VI Conclusion

In this study, we have developed a secure and energy-efficient QIoT-TMGO mechanism for the IoMT system through combining Quantum Key Distribution (QKD), trust-aware analytics, and MGO for dynamic clustering. Extensive simulation results demonstrate that the proposed model significantly outperforms eight state-of-the-art protocols in terms of key metrics. It particularly increases the network stability up to 24%, trust average scores up by 32% and obtains a data delivery ratio of 95.6% even with the presence of 10% malicious nodes infected into it. Moreover, the model further reaches First Node Death (FND) at 350 rounds, Half Node Death (HND) at 700 and Last Node Death (LND) at 1100 which demonstrates significant energy resilience. These findings show the framework to be reliable, robust and applicable for real-time communication of healthcare data even under adversarial circumstances. As future work, we will extend the model to mobility-aware scenarios and integrate blockchain-based audit trails for trust transparency. We also plan to validate performance on real hardware testbeds in clinical deployments.

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